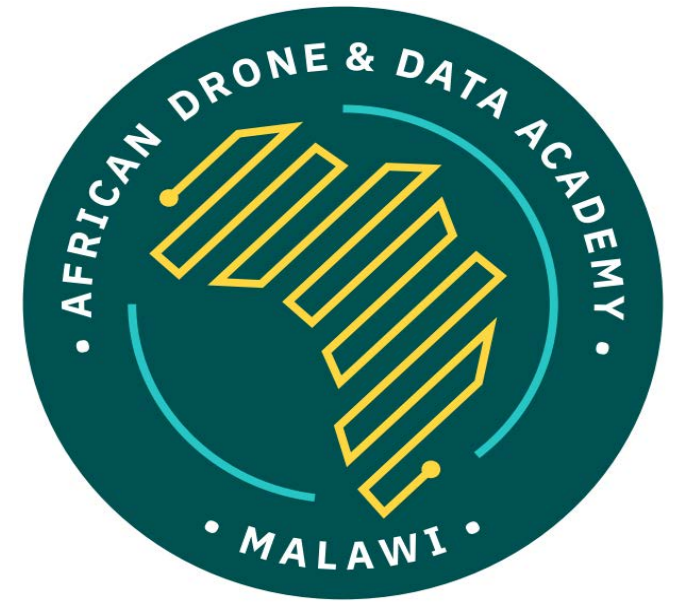




TOP Level 1

Day 3 Version A





TOP Level 1

Principles of Flight Part 2



VTOL aircraft can be analyzed as a system of three components: **structure, propulsion, control**

- Structure - central body and arm design
 - The loads are carried through joints in the central body, through the arms to the motors
 - The construction is simple and modular, making it easy to repair if damaged
 - It may be slightly heavier than a monocoque design, but its simplicity is appealing
 - The arms can be designed to fold for easy transport

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VTOL aircraft can be analyzed as a system of three components: **structure, propulsion, control**

- VTOL configurations
 - Quadcopters are a popular configuration due to cost and performance
 - Minimal motor and prop configuration reduces cost
 - Hexacopters are an optimal design for efficiency if load carrying capacity is a priority
 - ***Hexacopters are also redundant systems and if one motor fails, they will become a heavily loaded quadcopter and descend under control***
 - Motors can be located above and below the arms to create contra-rotating propulsion systems that greatly increase the load carrying capacity

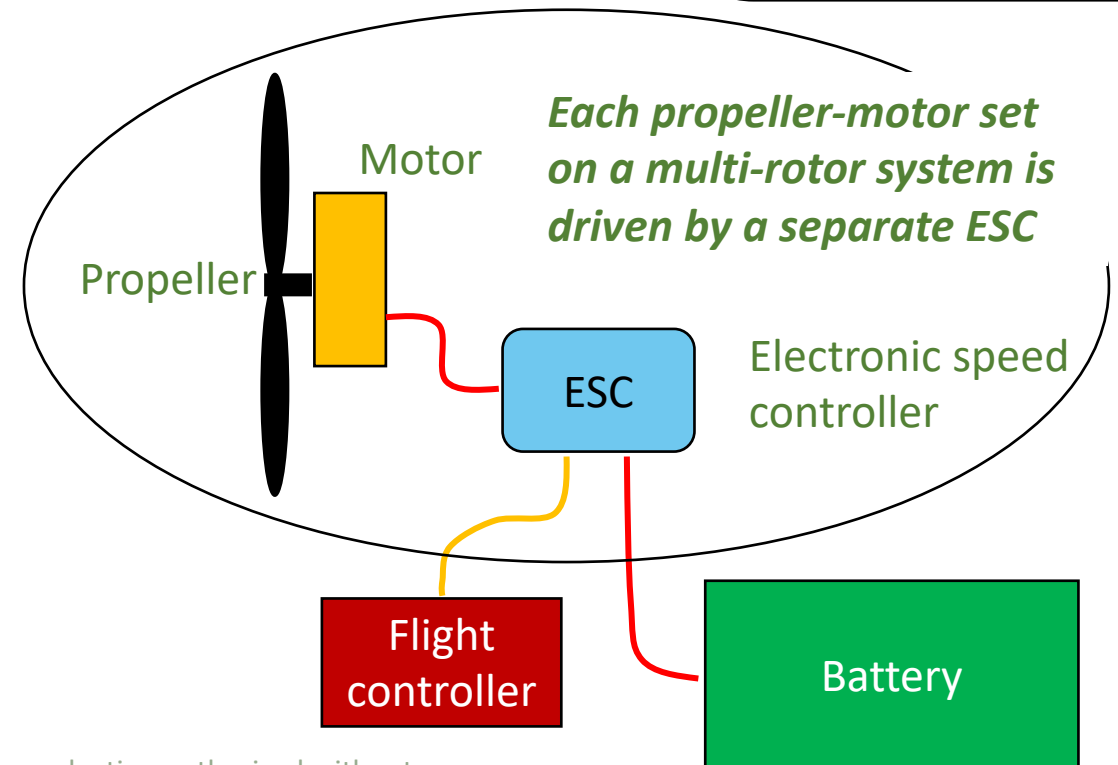
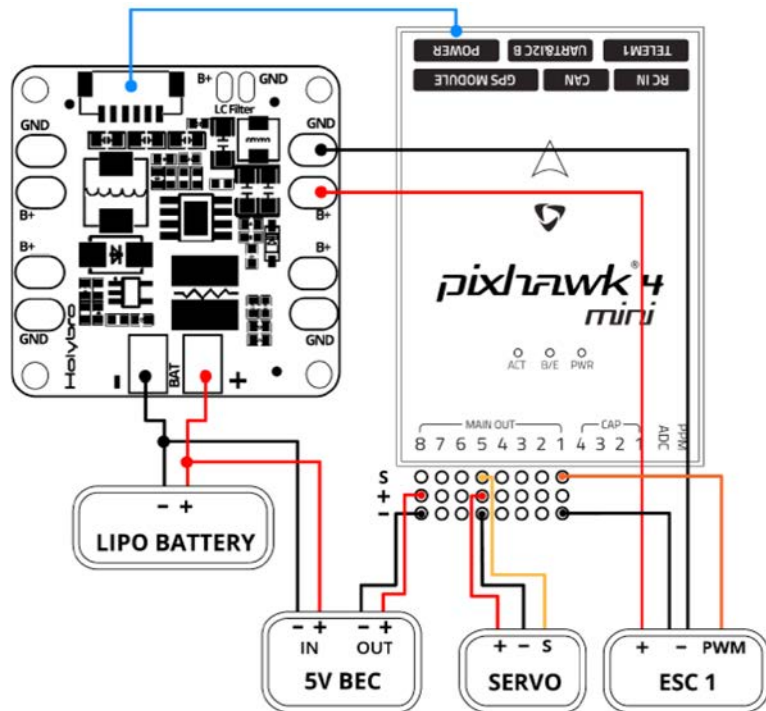
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VTOL aircraft can be analyzed as a system of three components: **structure, propulsion, control**

- The propulsion systems on VTOL aircraft require a flight controller that accepts a variety of motor/prop configurations

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VTOL propulsion: Motors

- Brushed and brushless DC motors work by energizing a series of electromagnets that are radially mounted within the motor body
- Combined with a series of magnets, a controlled magnetic force is created between the stator (fixed electromagnets) and the rotor (rotating outer shell with permanent magnets)
- Controlling the timing of the magnetic force results in a torque applied to the rotor

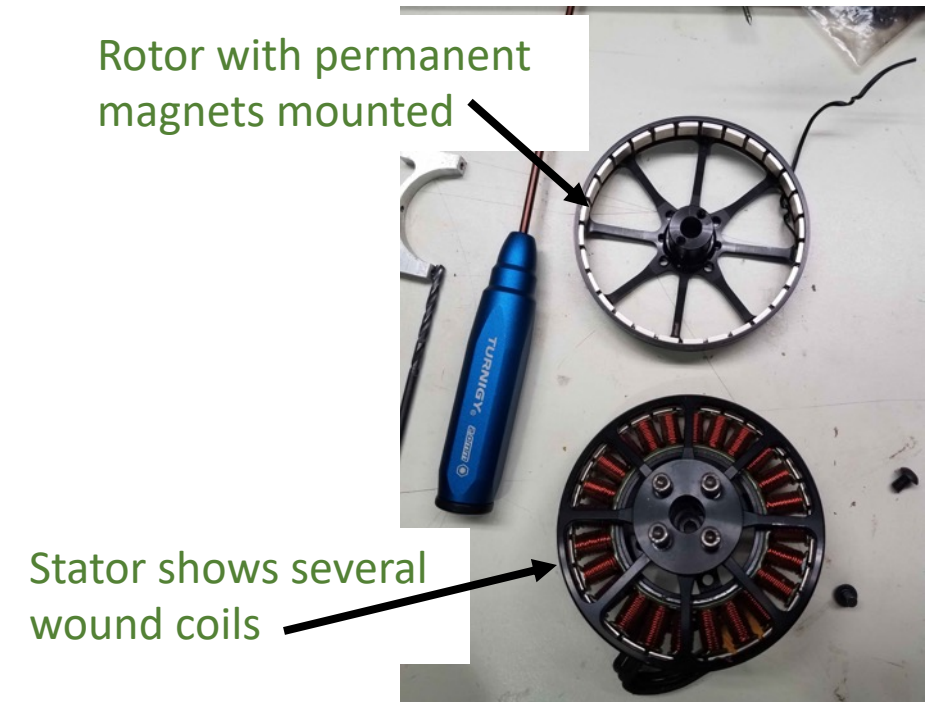
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VTOL propulsion: Motors

- Outrunner motors are simple and lightweight which has contributed to their popularity
 - In an outrunner motor, the magnets rotate (rotor) about the wound magnets (stator) which remain stationary
 - The stator magnets are controlled by the electronic speed controller (ESC)
 - Because of this configuration, the motor does not have brushes and it runs about 10% more efficient than a brushed motor version

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VTOL propulsion: Motors

- Outrunner motors also have the advantage of having many more poles than a typical brushed motor which gives them higher torque at lower RPM
 - This is better suited to the operating conditions for a motor/prop system
 - Motor performance may be described in terms of tested thrust as a function of power consumed
 - This represents a single datapoint of maximum thrust generated, whereas these motors will operate normally at much lower power settings
 - The propulsion data is also dependent on the selection or the propeller

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**The T-Motor
Antigravity 1005 is
a 255g motor that
claims to provide
11.5g thrust/Watt**



VTOL propulsion: Motors

- Common classifications of outrunner motors:
 - Some companies use a four digit reference to classify motors
 - A 2207 motor, for instance, would be 22mm in diameter with a 7mm stator height
 - **KV** rating of a motor is also used to define the speed at which the motor rotates
 - The KV value multiplied by the nominal voltage represents the theoretical no-load RPM of the motor
 - A 2300 KV motor running on a 2S battery would, for instance, turn at $\text{RPM} = 2300 \times 3.7 \times 2 = 17020 \text{ RPM}$

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The VT hex on the right uses T-motor 7005 KV230 motors, the S-500 on the left uses KV880 motors

VTOL propulsion: Motors

- The **KV** rating tells a lot about motor performance
 - The KV rating provides, most fundamentally, the rotational speed of the motor for a given applied voltage
 - KV also indicates the relative torque we can expect from the motor
 - A low KV motor has more windings of a thinner gauge wire which will carry fewer amps but at higher voltage, providing a higher torque output at low RPM
 - A high KV motor has fewer windings of a thicker gauge wire which will carry more amps at a lower voltage, turning the motor at a higher RPM with lower torque

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The S-500 has 2216 KV880 motors and uses a 4S battery

T-Motor performance chart for the 7005 KV230 that powers the VT hex

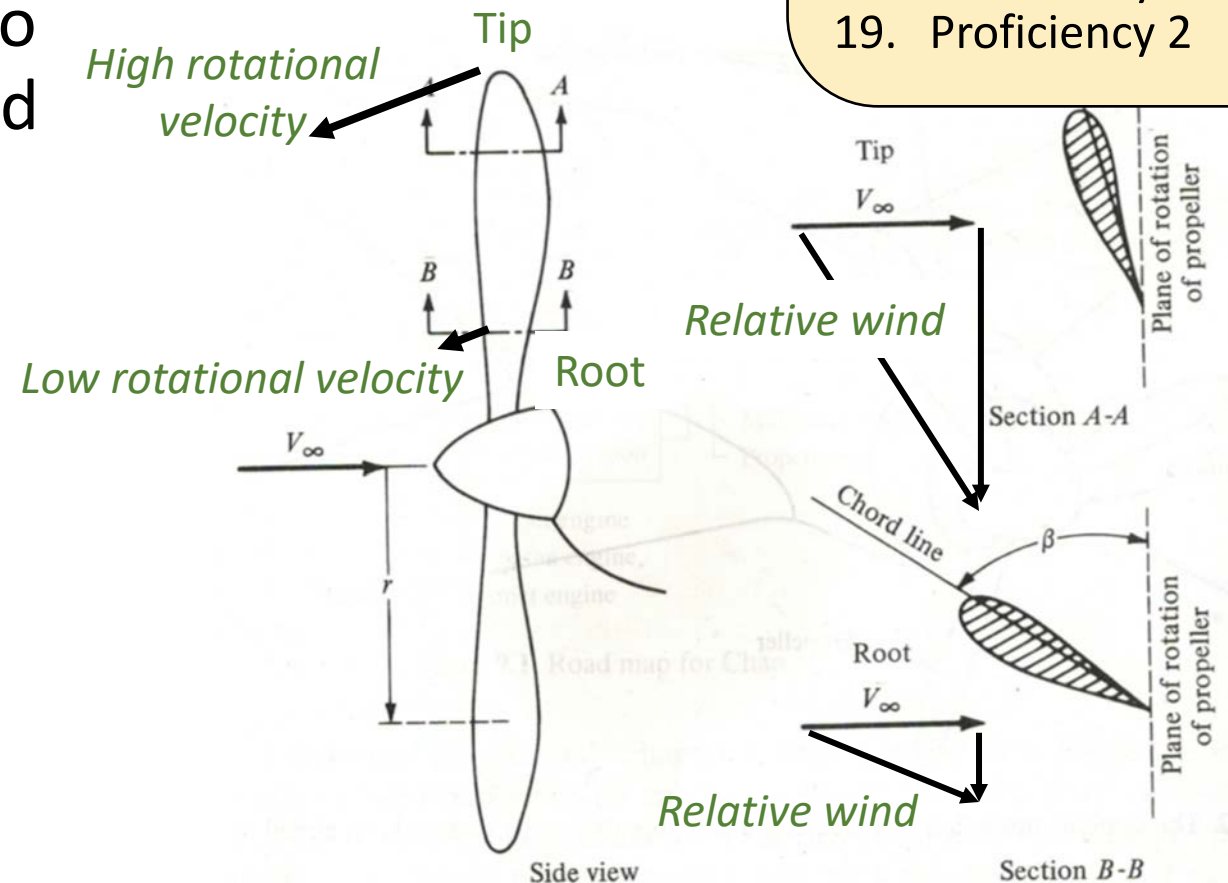
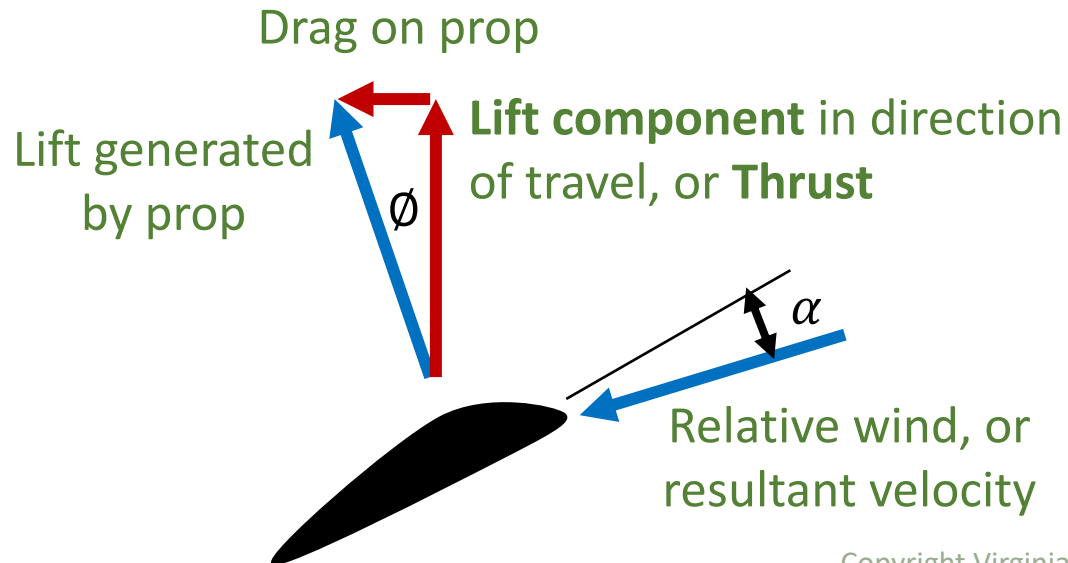
Item No.	Propeller	Throttle	Voltage (V)	Current (A)	Input power (W)	RPM	Torque (N*m)	Thrust (g)	Efficiency (g/W)	Operating Temperature
MN7005 KV230	T-motor P24*7.2CF	40%	23.79	4.40	104.60	2023	0.39	1407	13.44	64°C (Ambient temperature : 21°C)
		42%	23.78	4.79	113.88	2104	0.42	1501	13.17	
		44%	23.77	5.18	123.19	2165	0.43	1592	12.92	
		46%	23.75	5.71	135.63	2266	0.46	1698	12.51	
		48%	23.73	6.43	152.48	2347	0.49	1837	12.04	
		50%	23.71	7.04	167.01	2456	0.52	1951	11.68	
		52%	23.69	7.65	181.11	2531	0.54	2063	11.39	
		54%	23.67	8.18	193.69	2596	0.57	2153	11.11	
		56%	23.65	8.83	208.71	2684	0.60	2261	10.83	
		58%	23.62	9.53	225.12	2762	0.63	2378	10.56	
		60%	23.60	10.23	241.40	2835	0.65	2491	10.31	
		62%	23.58	10.96	258.50	2906	0.69	2605	10.07	
		64%	23.56	11.60	273.39	2967	0.72	2708	9.90	
		66%	23.52	12.64	297.29	3050	0.75	2841	9.55	
		68%	23.50	13.50	317.15	3120	0.78	2948	9.29	
		70%	23.47	14.31	335.87	3182	0.81	3060	9.10	
		75%	23.40	16.53	386.77	3335	0.89	3344	8.64	
		80%	23.33	18.79	438.46	3479	0.96	3632	8.28	
90%	23.16	23.77	550.60	3763	1.12	4184	7.59			
100%	22.99	29.15	670.20	4016	1.28	4691	6.99			
Note: Motor temperature is motor surface temperature @100% throttle running 10 mins (Data above based on benchtest are for reference only.Comparlson with that of other motor types is not recommended.)										

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VTOL propulsion: Propellers

- The twist that is built into a propeller keeps the angle of attack on the blade constant from root to tip, providing a uniform downward velocity to create thrust



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Propeller design considerations: size

- The size of a propeller is determined by many factors:
 - Dimension restrictions on the entire aircraft
 - The amount of weight that will be carried
 - The rotational speed of the propellers
- Larger diameter propellers turn slower because the tip speed of the propeller is $\mathbf{v_t = r\omega}$, where $\mathbf{r}$ is the radius on the propeller and $\mathbf{\omega}$ is the rotational speed.
- If we need to keep the tip speed below some maximum value, then the rotational speed must decrease as the diameter of the propeller increases
 - *Propeller tip speed that is too high results in more drag which in turn requires more power for the same thrust*

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Propeller design considerations: size

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Low efficiency hover:
small diameter propeller



Mid-efficiency hover: Large
hexacopter built by VT



High efficiency hover:
rotorcraft (Yamaha RMAX)

Transitional aircraft that both hover and fly forward are subject to inefficiencies both in hover and forward flight

- These aircraft are a compromise and are never flying at an efficient design point

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Quiz: *What compromises can you identify on this NASA aircraft design?*

In the selection of propellers, many factors are taken into consideration

- Carbon fiber propellers are strong and manufactured to tight tolerances, but they are expensive

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KDE propeller: \$120



T-motor 26x8.5 props: \$282/pair

In defining propellers, the first number is the diameter and the second number is the pitch. The pitch is the distance the propeller would move forward if there was no slip in air as it rotated

Carbon fiber propellers are usually manufactured by using a precision-machined mold to lay up the carbon epoxy material

- This is a good process for repeatability, but it represents an expensive process because it cannot be fully automated
 - Unlike plastic injection molding, hand lay-up is usually required to produce a carbon epoxy propeller
- A compromise can be found with carbon-reinforced plastics (CRP) which are injection molded (cheap) but strong (using carbon fiber mixed with plastic)

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<https://www.dolphinco.com.au/carbon-fiber-propeller/>

In the selection of propellers, many factors are taken into consideration

- Plastic propellers are low-cost but not as efficient because their shape is not well controlled (they are flexible)
- Wood propellers are able to hold better tolerance and still be low-cost

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EcoSoar propellers, 9x5, 5 for \$2.91

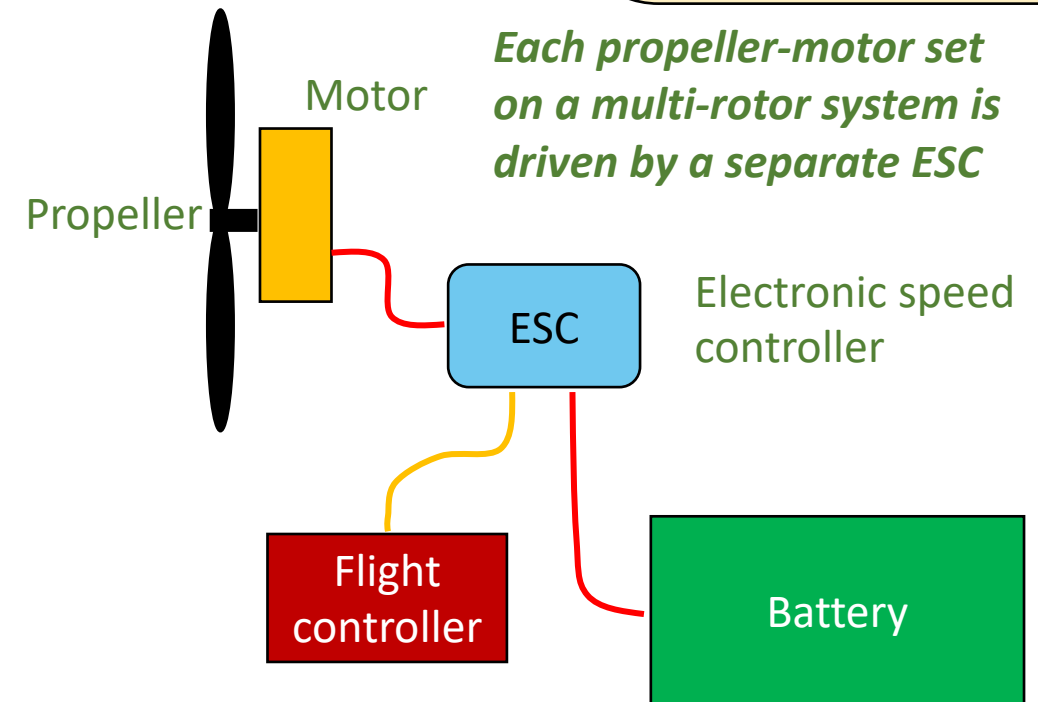


Gemfan 16x4.5, \$7.50

VTOL propulsion: Electronic speed controllers (ESC)

- The ESC manages the power from the battery using commands from the flight controller to precisely regulate the speed of the motors, either for fixed wing or VTOL flight
- The signal from the flight controller is a pulse-width modulated (PWM) signal typically 1ms to 2 ms to represent the full power range of the motor

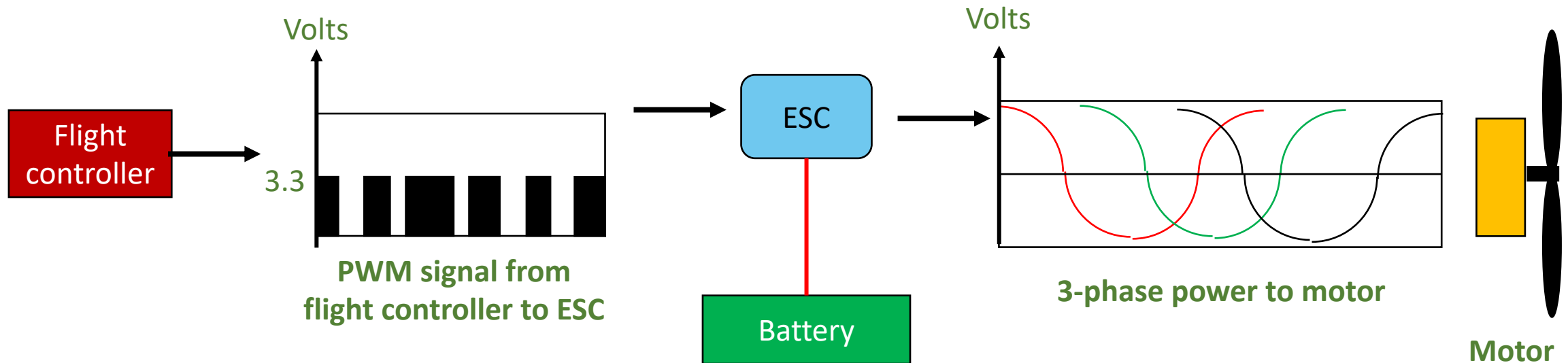
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VTOL propulsion: Electronic speed controllers (ESC)

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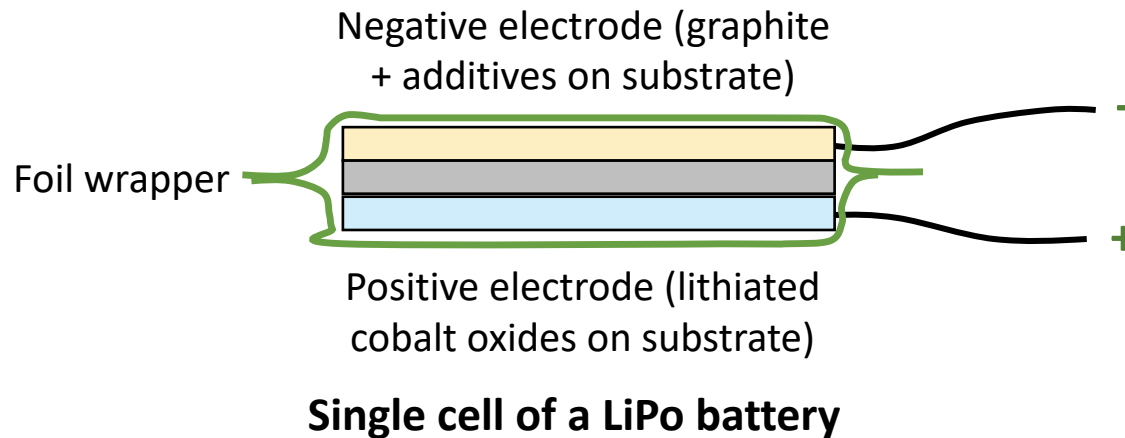
- The PWM signal to the ESC commands the ESC to send a three-phase variable frequency voltage to the motor



VTOL propulsion: Batteries

- Battery technology today is significantly better than just a few years ago, however with the advances in performance comes some risks
- The common battery used in drones is a Lithium Polymer (LiPo) battery design (typically 160Whr/kg)

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Each cell of a LiPo battery is contained in a soft aluminum pouch, allowing for battery expansion during charge and discharge cycles (up to 10% dimensional change)



A single LiPo battery cell has a nominal voltage of 3.7V

VTOL propulsion: Batteries

- A single LiPo cell will have a nominal (average voltage) of **3.7V (50% discharge)**
 - The lowest voltage that battery should experience is **3.0V**, and the highest voltage from charging should be **4.2V**
- A battery rated at **7.4V** is comprised of **two cells (2) wired in series (S)**
 - A **2S** battery has two cells in series (**7.4V**), a **3S** battery has 3 cells in series (**11.1V**)
- Batteries are also classified by their "C" rating - the EcoSoar battery is a 5200 mAh, 3S, 10 C battery

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VTOL propulsion: Batteries

- Note that batteries store **energy** which is in units of Watt-hours, or Whr.
- Lithium polymer batteries (most common for drones) have an **energy density** of about 160 Whr/kg
- The C rating indicates the maximum safe continuous discharge for the battery
- If the battery is a 10C battery and it is rated at 5200 mAhr (5.2Ahr), then the max safe continuous current would be $10 \times 5.2A = 52 \text{ Amps}$
- Some batteries also come with a burst rating, which would be the allowable current draw for short periods of time, like 10 sec

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TOP Level 1

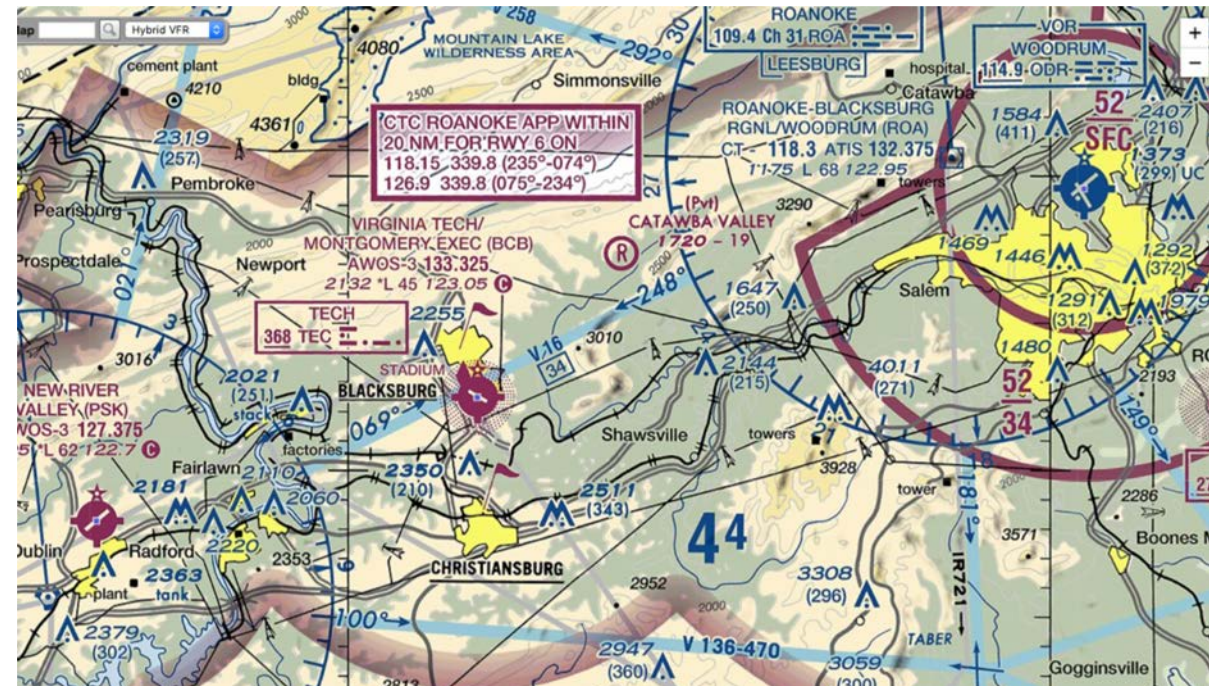
Navigational Theory



Navigation theory answers the question, “How do I get from point A to point B, and what will I encounter along the way?”

- The basic tool for navigation is the aeronautical chart
 - Aeronautical charts are published on a schedule so that you will always be updated when changes occur in the flying environment
 - They contain detailed information about airports, air routes, and terrain
 - They are used to check airspace prior to flight planning

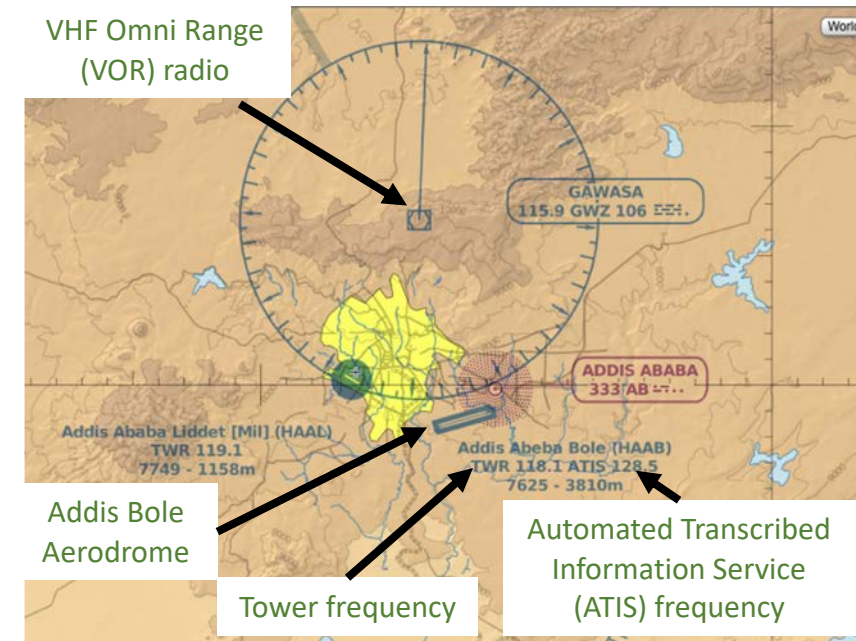
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During flight planning, consult charts to learn about aerodromes and airspace

- A visual flight rules (VFR) chart is useful to find landmarks when flying visually
- The chart also shows:
 - runway information
 - navigational aids
 - Restricted or prohibited areas
 - Populated areas
- Navigational aids are usually not used by RPA but they indicate where manned aircraft may be operating
- The frequencies that are listed on a chart are useful for monitoring traffic or recording current weather conditions from the ATIS broadcast
- Populated areas are shown in yellow which helps for flight planning purposes

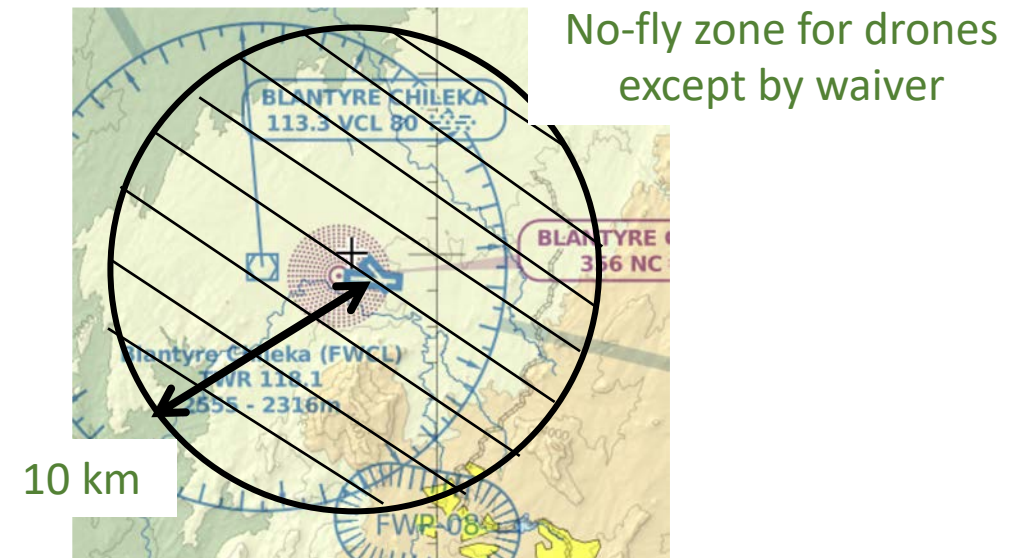
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During flight planning, consult charts to learn about aerodromes and airspace

- Drone operations should not be closer than 10 km to Class C - E aerodromes, or 7 km to Class A - B aerodromes
- Operations should not be near:
 - Approach or departure paths
 - Within the vicinity of navigational aids
 - Within the aerodrome traffic zone
 - Within the terminal traffic holding patterns
 - FWP or FWR areas may have drone restrictions - check with the authority

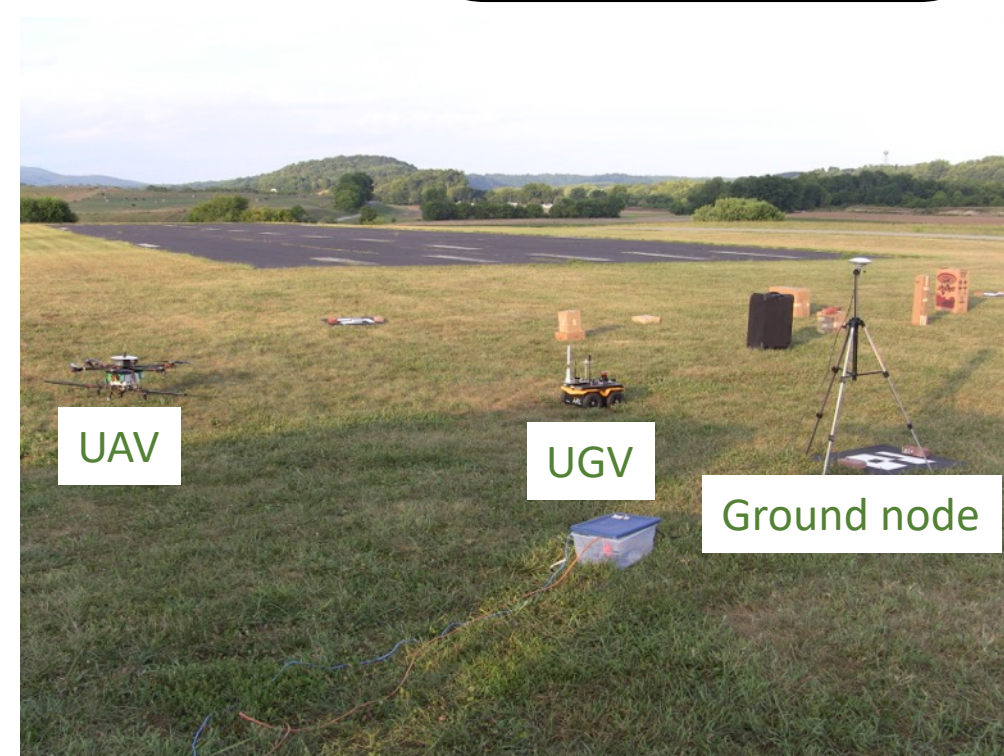
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Use of GPS in navigation

- GPS is the standard for RPA navigation and provides highly accurate information for precise positioning of the aircraft
 - Standard GPS will have an accuracy of a 1 - 2m RMS while systems that use a ground-based reference node can achieve centimeter-level accuracy
 - Systems with ground-based systems use a stationary ground receiver to compute local position changes of the moving node, which is more accurate than using a global-scale reference
 - Altitude accuracy is usually less than lateral and longitudinal accuracy and may be $\pm 3m$

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Use of GPS in navigation

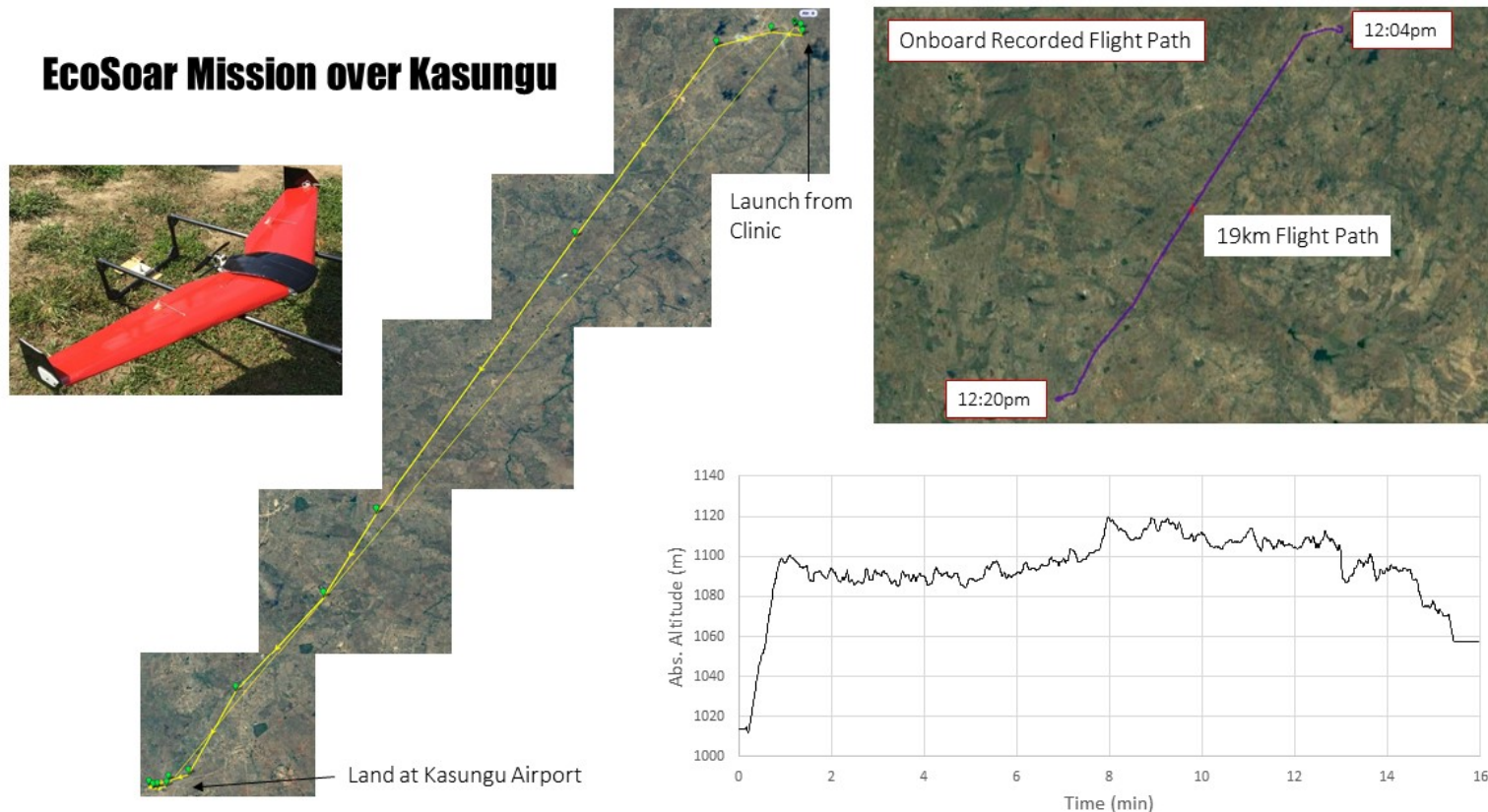
- Most flight controllers use a combination of position, heading and velocity information from the GPS and inertial measurement unit (IMU) for navigation
- If the GPS signal is degraded, the IMU will still provide heading information and the aircraft will approximately fly its intended course
 - The flight controller will usually alarm if the number of satellites drops below a minimum number
 - A GPS nav system will have significantly more error when receiving 5 satellites as opposed to 12
 - If the number of satellites received drops below the minimum required for navigation, the aircraft failsafe mode will command an immediate landing

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GPS errors will be bounded at a constant value, allowing an accurate review of flight track data

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Flight logs provide a valuable source of information for analyzing flight performance or accident data

It is also possible to use other sensed data, such as visual data, to perform visual servoing for flight control

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Nadir imagery captured by aircraft

